

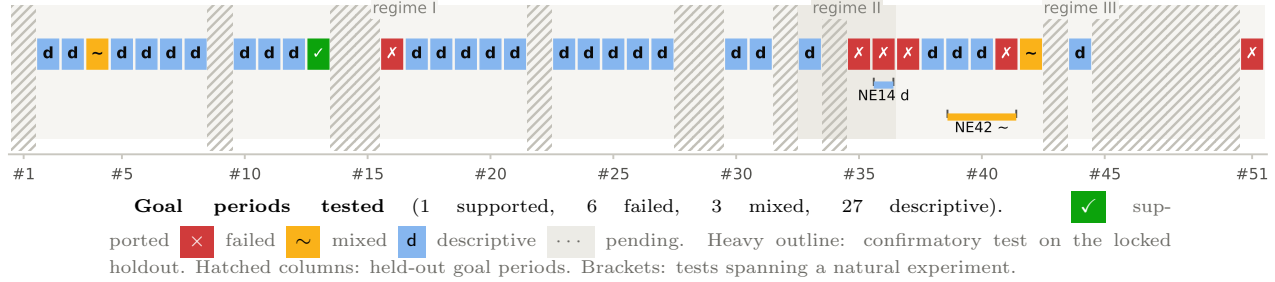
H99: Fluctuations predict relaxation (mean-field Glauber)

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 2/5 Usefulness 2/5

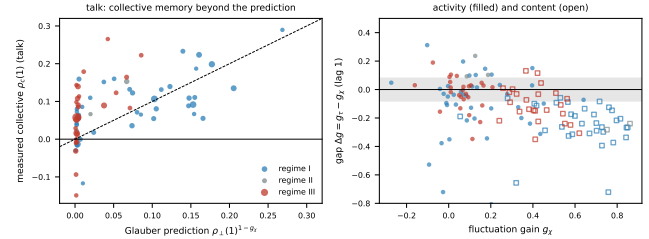
Bottom line. Equal-time fluctuations do not predict the swarm's relaxation where talk coupling is real: the collective mode outlives the mean-field Glauber prediction (delayed read-out coupling or a slow conversation field), kicks outlive it tenfold, and content behaves as a field.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

Mean-field Glauber dynamics ties two numbers together: the gain g read from equal-time fluctuations must also set how much slower the collective mode relaxes than one agent. HH81 asked whether this holds without inferring J . The test covers 70 non-holdout units in 35 periods, on talk, activity (control) and content. Talk minutes carry no single-agent memory ($\rho_{\perp}(1) \approx 0.006$), so a post hoc statistic, the collective memory excess $\Delta\rho_1$, replaced the planned log-ratio. **Where talk coupling is real (regime III), the collective outlives the prediction in every resolved period:** $\Delta\rho_1 = +0.067 [0.034, 0.100]$ in #51. Regime I passes, but the test has no power there. Content is a field. Kicks outlive the fluctuation clock: the #51 nudge response is $\times 9.5$ the Onsager value. The holdout is not run.



Observables. Left: each point is one talk unit (size $\propto$ trimmed minutes). The x-axis is the Glauber prediction for the collective lag-1 memory, the y-axis the measured value. Regime-III units (red) sit above the diagonal: the swarm remembers talk longer than its equal-time gain allows. Right: activity (filled) and content (open) gap Δg against g_{χ} ; content lies below the band, the field-like side.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For a monitor: the equal-time talk dial is not a distance to criticality in regime III. There the collective talk mode keeps memory for minutes ($\Delta\rho_1 \approx 0.07$) that no agent has on its own, which is what a coupling delayed by one read-out call produces. Use a lagged estimator (H67's g_{lag}) and report $\Delta\rho_1$ beside the dial as a check. For an operator: a nudge's effect does not decay like spontaneous activity. Expect about ten times more response, spread over 10–30 min, than the agent's own one-minute clock predicts, and do not read the kick kernel as a susceptibility. In regime I nothing here distinguishes a coupling from a fast shared field.

PHYSICAL MODEL

Models 02 and 01. Spin $s_i(t)$: agent i talks (or acts) in minute t , centred within 30-min blocks on the all-present window; content uses 32-d style-residualized window vectors. M_t is the collective (uniform) mode, $X_t^{\perp}$ the transverse modes. Linearized mean-field Glauber relaxes M at rate $(1-g)/\tau_0$ and $X^{\perp}$ at $1/\tau_0$, with the same $g = \beta J_0(1-m^2)$ that sets the equal-time variance ratio.

$$\tau_0 \dot{m} = -m + \tanh(\beta(J_0 m + h)) \Rightarrow \tau_c = \tau_0/(1-g)$$

$$g_{\chi} = 1 - \sum_t p_t / \sum_t c_t, \quad c_t = M_t^2, \quad p_t = (|X_t|^2 - M_t^2)/(N-1)$$

$$\Delta g = \left[1 - \frac{\ln \rho_c(1)}{\ln \rho_{\perp}(1)}\right] - g_{\chi}, \quad \Delta \rho_k = \rho_c(k) - \rho_{\perp}(k)^{1-g_{\chi}}$$

H99-glauber-fluctuation-relaxation · generated 2026-10-04 14:04 by in-fra/summaries/build_summaries.py · card hypotheses/H99-glauber-fluctuation-relaxation/README.md · ratings: H99 agent (suggested)

RESULTS VS PREDICTIONS

Prediction (dated)	Observed	Verdict
S1–S6 synthetic (real masks)	Glauber consistent 78–100%; fast field 78% ($\tau_0=1$ min); slow field only at lag 2	3/6 pass
A2 power ($\tau_0=15$ s)	slow field 88–100%, delay 100%; limits P2 fast field 0–6%	
P1 regime-III talk consistent	0/4 resolved periods; 8 slow vs 7 failed consistent units; $\Delta\rho_1 +0.077$	failed
P2 regime-I talk = fast field	2/25 fast calls; no power	not testable
P3 activity inconsistent $\geq 50\%$	9/16 resolved periods (fast, regime I)	supported
P4 content $\Delta g \geq 0$	$\Delta g_1 < 0$ in 92% of 60 units (median -0.17)	failed (sign)
P5 kick decay Glauber	$\approx \#4c \lambda 0.54 [0.21, 0.86]$ vs 0.15; #51 nudge $\times 9.5$	failed
P6 g_χ upper bound 0.6	all but 3 small regime-I units	mostly
Natives NE42, NE14, G51	merge collapses g_χ ($-0.13, -0.22$); 36c slow; nudge tail	mixed / failed

GOAL PERIODS IN DETAIL

G02 d descriptive
G03 d descriptive
G04 $\sim$ mixed
G05 d descriptive
G06 d descriptive
G07 d descriptive
G08 d descriptive
G10 d descriptive
G11 d descriptive
G12 d descriptive
+27 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The talk statistic and its rule were chosen after the talk autocorrelations were seen (A2, post hoc). The one-minute bin is coarser than the talk clock, so regime I is untestable and g_τ is undefined there. A slow conversation field and a delayed coupling give the same sign; only the match with H67's read-out gain points to coupling. Content keeps the day's topic field (day centring faked anti-correlation). The nudge comparison uses H04's kernel, whose tail H59 attributes to the nudger's selection.

ROUND 2

Re-index talk on the per-call clock and fit an instantaneous-plus-one-call kernel; align human-message kicks on the receiving call; confirm C1–C4 on #22, #28, #43.

- **Essence.** a dynamical check that tells a coupling from a field without inferring J, on the clock at which agents actually update.
- **H99-R1** Per-call clock.
- **H99-R2** Two-timescale kernel.
- **H99-R3** Regime-I power.
- **H99-R4** Kicks on the receiving call.
- **H99-R5** Content fields.

Second figure. Synthetic validation on real trimmed masks. Left (A2, agent clock 15 s): $\Delta\rho_1$ (blue) and $\Delta\rho_2$ (orange) are near 0 in independent and Glauber worlds, positive for slow fields and delayed coupling, negative only for a strong fast field. Right (A1, clock 1 min): the lag-1 and lag-2 gaps Δg . Grey bands are the decision thresholds.

